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PALAEOMAGNETISM AND K-Ar DATING OF THE CARBONIFEROUS ANDACOLLO SERIES (ARGENTINA) AND THE AGE OF ITS HYDROTHERMAL OVERPRINTING

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Palaeomagnetic data from 15 hand samples collected in a sequence of tuffs of Carboniferous age (Andacollo Series) exposed in western Argentina (37°10'S, 70°38'W) are given. These samples show hydrothermal alteration. After AF and thermal cleaning the majority of samples show reversed polarity and yield a palaeomagnetic pole at 73.6°S, 64.7°E ($d\psi = 6°$, $d\chi = 9°$). By comparing the position of this pole with the apparent polar wander curve recognized for the Late Palaeozoic and Mesozoic of South America, quantitative support for a Cretaceous age for the cleaned remanence of these samples is given.

K-Ar dating of the Andacollo rocks indicates a mean age of 69 ± 4 m.y.; this suggests that the isotopic system of these rocks became closed in Late Cretaceous time.

Palaeomagnetic and radiometric data agree indicating a Cretaceous age for the hydrothermal alteration of the Andacollo rocks.

1. Introduction

The palaeomagnetic group of the Universidad de Buenos Aires has programmed the study of Palaeozoic rocks older than Late Carboniferous of South America. In this programme are included the Carboniferous tuffs from the Andacollo Series (Province of Neuquén, western Argentina). During the field work we noted that the Andacollo samples we collected shown evidence of hydrothermal alteration. Therefore, we collected a few of samples for preliminary palaeomagnetic study to see whether their natural remanent magnetization was acquired in Carboniferous time or later during the period of alteration (secondary chemical or/and thermal remanent magnetization). We present here the results of this palaeomagnetic study and of the whole rock K-Ar age determinations carried out on some of the Andacollo samples.

2. Geological evidence

In the western part of Argentina (37°10'S, 70°38'W, Province of Neuquén) continental deposits and tuffs included in the Andacollo Series are exposed. These rocks were formed in Carboniferous time and have been classified in three formations named in chronological order: Tobas Inferiores, Huaraco Formation and Tobas Superiores [1]. This palaeomagnetic-radiometric study includes a collection of hand samples from the Tobas Inferiores.

The western part of Neuquén was affected by magmatic episodes (Permian, Triassic, Jurassic, and Cenozoic), reiterative epeirogenetic episodes of subsidence (Lias, Tithonian and late Senonian) and uplift (Kimmeridgian, Neocomian and Eocene) and tectonic movements (Senonian). Finally, this part of Neuquén was affected by an orogenic episode (Andean movements) during the Cenozoic [1,2].

The Andacollo rocks included in this study show hydrothermal alteration (propylitization-silicification and, to a lesser degree, argillization and sericitization).

3. Palaeomagnetic study

Fifteen hand samples of tuffaceous rocks were collected from the Tobas Inferiores of the Andacollo

Series exposed along the route between Andacollo and Huingancó Village. Samples were collected at different stratigraphic levels in three sites and oriented using both magnetic and sun compasses.

Forty-five cylinders of 2.5 cm diameter and height were cut. A Digico spinner magnetometer was used to measure the remanent magnetization of the cylinders. Detailed alternating field and thermal demagnetization were used to isolate the stable remanence of two different cylinders of each sample [3]. A third cylinder from the same sample was submitted to the optimum alternating field so as to improve the definition of the direction of the stable remanence of each sample.

A weak secondary component of remanence was present in the majority of the samples; this secondary remanence has the characteristics of a viscous remanent magnetization and was destroyed by the alternating field and thermal demagnetization. The optimum demagnetizing AC field varied between 100 and 150 Oe at which more than 80% of the natural remanent magnetization (NRM) was retained; the optimum demag

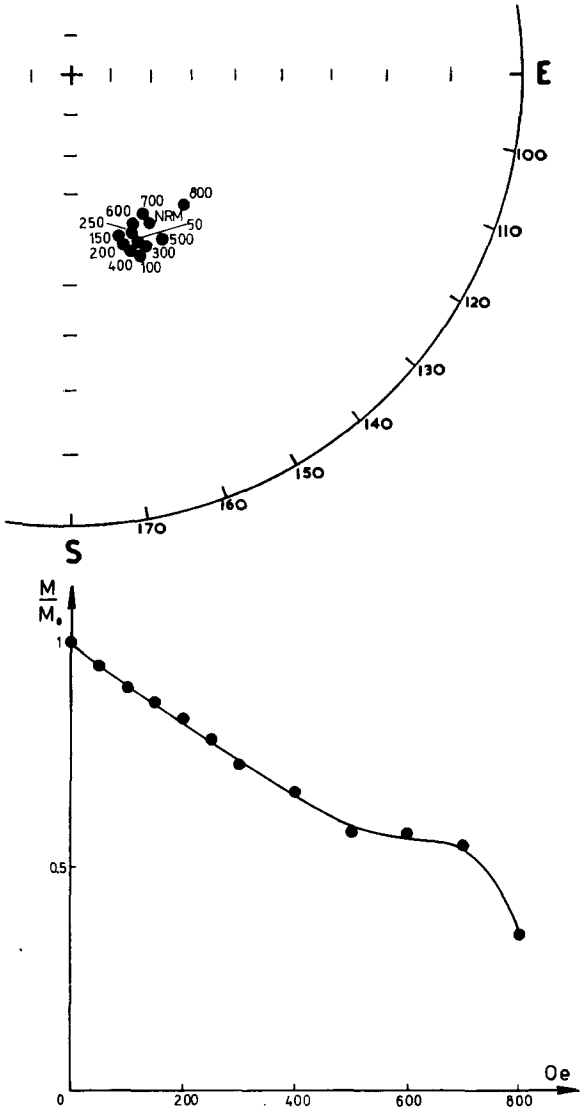


Fig. 1. Changes in direction and intensity of the natural remanent magnetization of one specimen from the Andacollo Series in response to AF treatment.

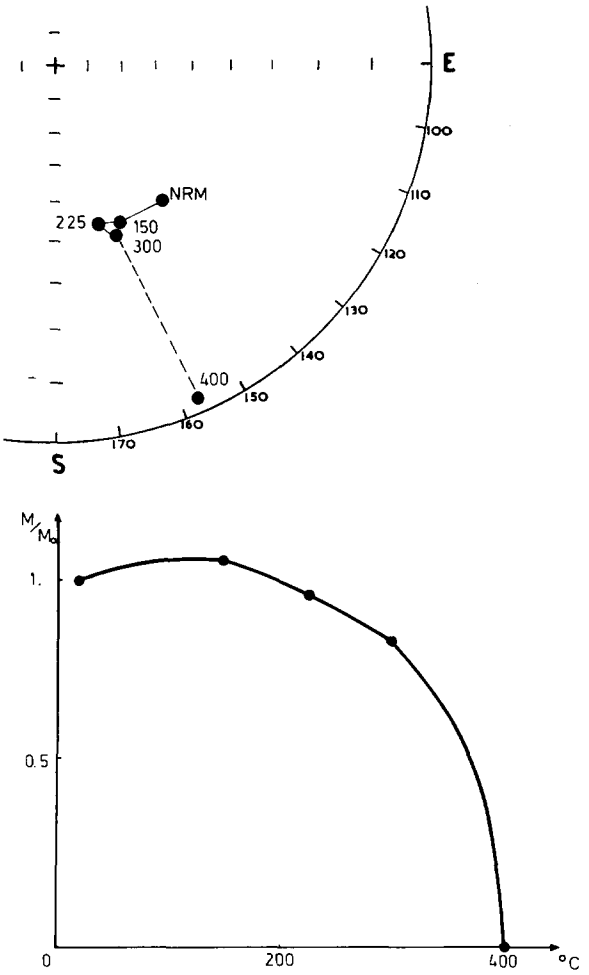


Fig. 2. Changes in direction and intensity of the NRM of one specimen of the same sample from the Andacollo Series as in Fig. 1, in response to thermal treatment.

TABLE 1

Summary of palaeomagnetic results of the tuffaceous rocks of the Andacollo Series (Tobas Inferiores) exposed in western Argentina (Andacollo-Huingancó zone, Province of Neuquén

Natural remanent magnetization							Cleaned remanent magnetization							Palaeomagnetic pole			
referred to present horizontal					referred to palaeohorizontal		referred to present horizontal					referred to palaeohorizontal					
N	k	α_{95} (°)	D (°)	I (°)	D (°)	I (°)	N	k	α_{95} (°)	D (°)	I (°)	D (°)	I (°)	lat (°)S	long (°)E	$d\chi$ (°)	$d\psi$ (°)
13	31.8	7.4	208.8	78.3	165.2	45.8	13	30.2	7.6	204.7	75.0	167.4	42.8	73.6	64.7	9	6

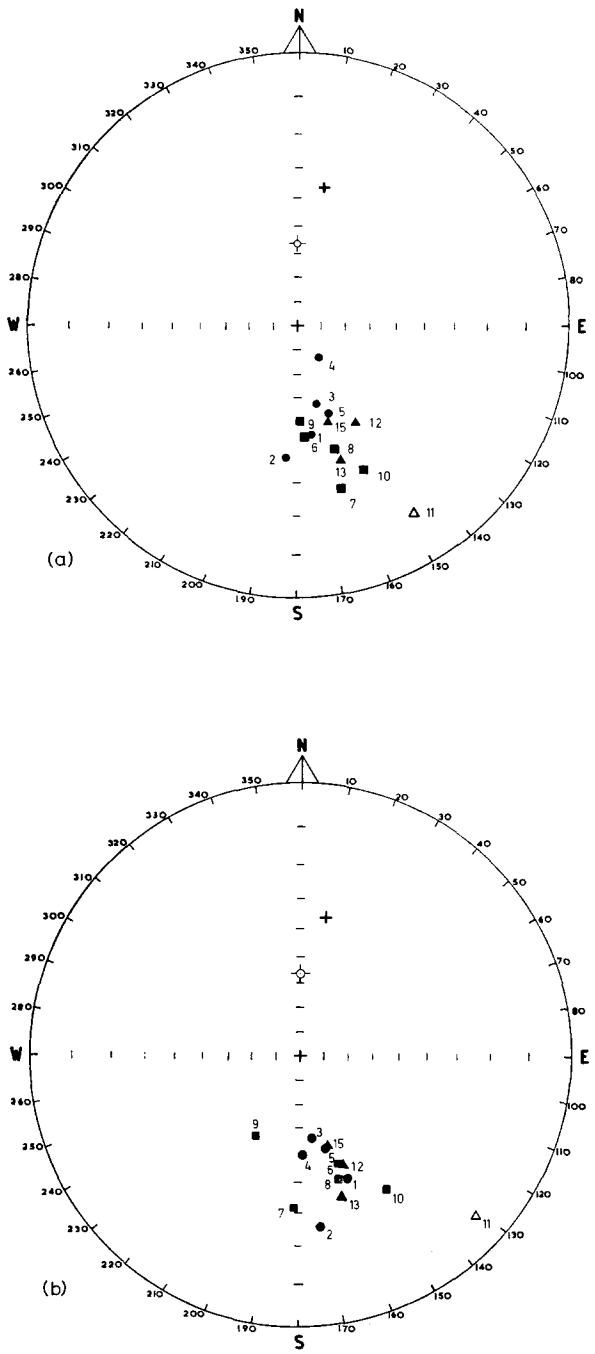


Fig. 3. Andacollo Series. Directions of (a) NRM and (b) cleaned remanent magnetization of hand samples referred to the palaeohorizontal. Solid symbols indicate downward dipping directions. The directions of the present geomagnetic field and the axial dipole are shown by + and ⊹ respectively.

netizing temperature varied between 150° and 300°C at which more than 70% of the natural remanence intensity was retained. The NRM was wholly destroyed after thermal treatment at about 400°C; samples have blocking temperatures from 300° to 400°C. Figs. 1 and 2 show the changes in direction and intensity of NRM of specimens from one hand sample (number 9012) in response to AF and thermal treatments, respectively. It was not possible to isolate stable remanence in one sample.

The method of Fisher [4] was used to compute the mean directions of natural and cleaned remanent magnetization of each sample assigning unit weight to the direction of each of its cylinders. Sample-mean directions of NRM and cleaned remanent magnetization were referred to the present and palaeohorizontal; however, the fold test could not be applied because all the samples were collected from an area where the dip of the strata is uniform (dip, 37°S; strike 61°E).

The sample-mean directions are reasonably well grouped before as well as after cleaning (Fig. 3a and b, respectively). The cleaned direction of sample 9011 does not lie in the main group of reversely polarized directions; we associate this oblique direction to a geomagnetic excursion or a polarity transition of the geomagnetic field. The statistical method of Fisher was used to compute the mean directions of NRM and cleaned remanence for the Tobas Inferiores of the Andacollo Series assigning unit weight to the mean direction of each hand sample (Table 1); we have not included sample 9011 in this mean. A palaeomagnetic pole was calculated for the Andacollo Series using the mean direction of cleaned remanence referred to the palaeohorizontal; the coordinates of this pole are given in Table 1.

4. Potassium-argon age determinations

Whole K-Ar age determinations were carried out on three hand samples of the Andacollo Series. The study was carried out in the Instituto de Geocronología y Geología Isotópica (Universidad de Buenos Aires). The techniques used are those indicated by Linares and Valencio [5]. The K-Ar ages are shown in Table 2; the mean of the ages is 69 ± 4 m.y., suggesting a Late Cretaceous age.

TABLE 2

K-Ar ages for tuffs of the Andacollo Series (Tobas Inferiores) exposed along the route between Andacollo and Huinganco Village, Province of Neuquén, Argentina

Sample No.	K (%)	$^{40}Ar_{rad}$ (10^{-10} mol/g)	$^{40}Ar_{atm}$ (%)	Age (m.y.)
9001	5.42	6.57	10.7	67 ± 3
9006	2.79	3.71	24.8	73 ± 2
9006	2.79	3.76	9.9	75 ± 2
9010	4.48	5.25	23.6	65 ± 3

5. Interpretation of results

The reversely polarized directions of the natural and cleaned remanences, the small change in the directions of remanence after AF and thermal treatment and the clustering of the natural and cleaned remanence directions indicate that the samples' natural remanence is stable. The carriers of this remanence are magnetic minerals with blocking temperatures from about 300° to 400°C.

The mean direction of the cleaned remanence of the Andacollo rocks is not consistent with the direction of the Carboniferous geomagnetic field in South America. Fig. 4a shows that the position of the palaeomagnetic pole for the Andacollo Series (SAK_9) lies far away from the mean position for the Stephanian palaeomagnetic poles for South America (SACu [6]) and the position of the pole for the Moscovian Taiguati Formation (SAC_2 [7]). The position of the Andacollo pole is close to the Cretaceous palaeomagnetic poles for South America (Fig. 4a, b [8]); this suggests a Cretaceous age for the isolated stable remanence of the Andacollo rocks. The carriers of this remanence can be either the original magnetic minerals of the Andacollo rocks or magnetic minerals formed during the period of hydrothermal alteration. In the first case the stable remanence would be a secondary thermal remanent magnetization (TRM) acquired when the Andacollo rocks cooled in

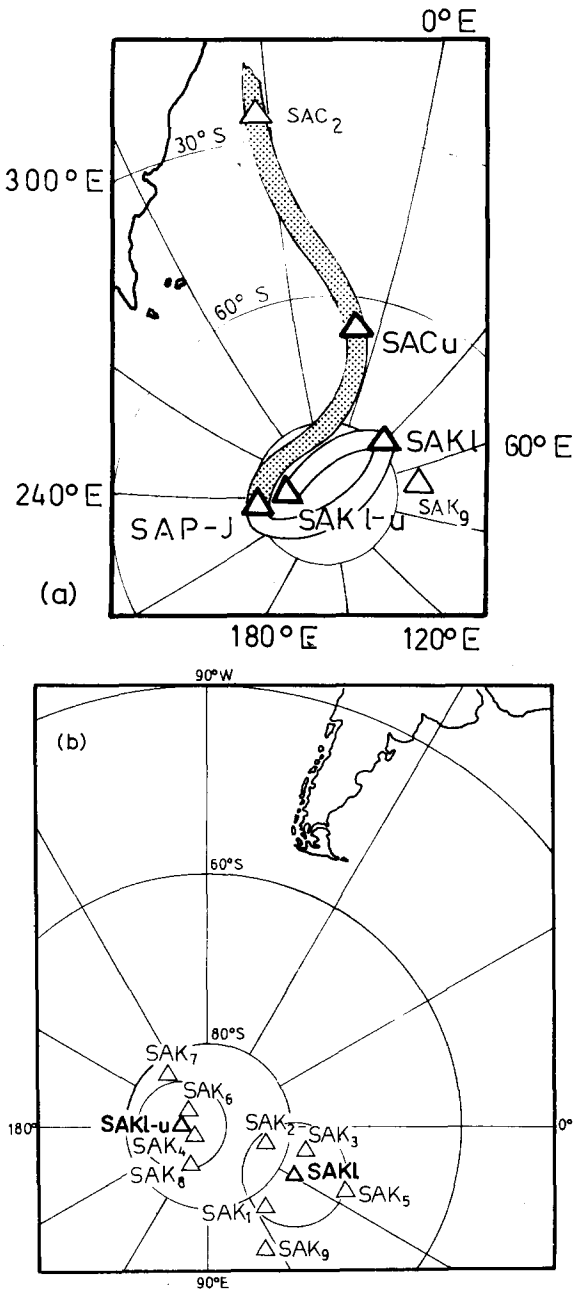


Fig. 4. (a) Late Palaeozoic–Mesozoic section of the apparent polar wandering curve and (b) Cretaceous palaeomagnetic poles for South America. SAK_9 is the palaeomagnetic pole for the Andacollo Series. The other Cretaceous palaeomagnetic poles are: SAK_1 = Serra Geral Formation, 115–125 m.y.; SAK_2 = Vulcanitas Cerro Colorado Formation, 121 ± 3 m.y.; SAK_3 = Rio de los Molinos dykes, 138 m.y.; SAK_4 = Vulcanitas Rumipallla Formation, <121 m.y.; SAK_5 = Vulcanitas de Almafuerte, 123 m.y.; SAK_6 = Pirgua Subgroup, 114–77 m.y.; SAK_7 = Poços de Caldas Alkaline Complex, 75 m.y.; and SAK_8 = Serra Geral Formation [8]. SAP-J is the mean position of the quasi-static period defined by the Permian, Triassic and Jurassic poles [8]. Other references are given in the text.

presence of the Cretaceous geomagnetic field at the end of the period of alteration. In the second case the stable remanence would be a secondary chemical remanent magnetization (CRM) acquired during the period of alteration in Cretaceous time.

The K-Ar age determinations carried out on Andacollo rocks indicate a Late Cretaceous age; this indicates that the isotopic system of these rocks became closed when they cooled in Late Cretaceous time (69 ± 4 m.y.).

If the stable remanence of the Andacollo rocks is a TRM, the palaeomagnetic and radiometric results are dating the episode of cooling subsequent to the period of hydrothermal alteration. If it is a CRM the palaeomagnetic results date the period of alteration and the radiometric results date the period of cooling posterior to it.

To proceed further, Fig. 4b shows that the position of SAK_9 lies closer to the mean position of the Early Cretaceous palaeomagnetic poles for South America (SAK1) than to the mean position of the quasi-static period defined by the late Early and Late Cretaceous poles (SAK1-u). This suggests that the Andacollo rocks acquired their stable remanent magnetization a little earlier than when their isotopic system became closed and supports a chemical origin for the remanence.

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